

ALFAAS60322

## Silver

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:** 银  
**Product Description:** Silver

**Cat No. :** S60322  
**CAS No** 7440-22-4  
**Molecular Formula** Ag

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Solid

**Appearance**  
Silver

**Odor**  
Characteristic

**Emergency Overview**  
Very toxic to aquatic life with long lasting effects.

#### Classification of the substance or mixture

|                          |            |
|--------------------------|------------|
| Acute aquatic toxicity   | Category 1 |
| Chronic aquatic toxicity | Category 1 |

#### Label Elements



**Signal Word**

**Warning**

#### Hazard Statements

H410 - Very toxic to aquatic life with long lasting effects

**Precautionary Statements****Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

The product contains no substances which at their given concentration are considered to be hazardous to health.

**Environmental hazards**

Very toxic to aquatic life with long lasting effects. Is not likely mobile in the environment due its low water solubility. Spillage unlikely to penetrate soil.

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component | CAS No    | Weight % |
|-----------|-----------|----------|
| Silver    | 7440-22-4 | <=100    |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

None reasonably foreseeable.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

approved class D extinguishers.

**Extinguishing media which must not be used for safety reasons**

Water may be ineffective.

**Specific Hazards Arising from the Chemical**

Do not allow run-off from fire-fighting to enter drains or water courses.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### SECTION 6. ACCIDENTAL RELEASE MEASURES

#### Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

#### Environmental Precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Should not be released into the environment.

#### Methods for Containment and Clean Up

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

### SECTION 7. HANDLING AND STORAGE

#### Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

#### Storage

Protect from moisture. Keep away from acids.

#### Specific Use(s)

Use in laboratories

### SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

#### Control Parameters

| Component | China | Taiwan                      | Thailand | Hong Kong                  |
|-----------|-------|-----------------------------|----------|----------------------------|
| Silver    | -     | TWA: 0.01 mg/m <sup>3</sup> |          | TWA: 0.1 mg/m <sup>3</sup> |

| Component | ACGIH TLV                  | OSHA PEL                                                             | NIOSH                                                                                   | The United Kingdom                                                    | European Union                  |
|-----------|----------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------|
| Silver    | TWA: 0.1 mg/m <sup>3</sup> | (Vacated) TWA: 0.01 mg/m <sup>3</sup><br>TWA: 0.01 mg/m <sup>3</sup> | IDLH: 10 mg/m <sup>3</sup><br>TWA: 0.01 mg/m <sup>3</sup><br>TWA: 0.9 µg/m <sup>3</sup> | STEL: 0.3 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr | TWA: 0.1 mg/m <sup>3</sup> (8h) |

#### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

#### Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust MDHS 91 Metals and metalloids in workplace air by X-ray fluorescence spectrometry MDHS 99 Metals in air by ICP-AES

#### Exposure Controls

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**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

|                                 |                          |                                          |
|---------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>               | Silver                   |                                          |
| <b>Physical State</b>           | Solid                    |                                          |
| <b>Odor</b>                     | Characteristic           |                                          |
| <b>Odor Threshold</b>           | No data available        |                                          |
| <b>pH</b>                       | No information available |                                          |
| <b>Melting Point/Range</b>      | 961.8 °C / 1763.2 °F     |                                          |
| <b>Softening Point</b>          | No data available        |                                          |
| <b>Boiling Point/Range</b>      | 2210 °C / 4010 °F        |                                          |
| <b>Flash Point</b>              | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>         | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b> | No information available |                                          |
| <b>Explosion Limits</b>         | No data available        |                                          |

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|                                         |                          |         |
|-----------------------------------------|--------------------------|---------|
| Vapor Pressure                          | No data available        |         |
| Vapor Density                           | Not applicable           | Solid   |
| Specific Gravity / Density              | 10.5 g/cm3               | @ 20 °C |
| Bulk Density                            | No data available        |         |
| Water Solubility                        | Insoluble in water       |         |
| Solubility in other solvents            | No information available |         |
| Partition Coefficient (n-octanol/water) |                          |         |
| Autoignition Temperature                | No data available        |         |
| Decomposition Temperature               | No data available        |         |
| Viscosity                               | Not applicable           | Solid   |
| Explosive Properties                    | No information available |         |
| Oxidizing Properties                    | No information available |         |

Molecular Formula Ag

## SECTION 10. STABILITY AND REACTIVITY

|                          |                                              |
|--------------------------|----------------------------------------------|
| Stability                | Stable under recommended storage conditions. |
| Hazardous Reactions      | None under normal processing.                |
| Hazardous Polymerization | No information available.                    |
| Conditions to Avoid      | None known.                                  |
| Materials to avoid       | Acids. Oxidizing agent.                      |

Hazardous Decomposition Products Silver oxides.

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation              |
|-----------|---------------------------|---------------------------|------------------------------|
| Silver    | LD50 > 5000 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( rat ) | LC50 > 5.16 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

## (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

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(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

Symptoms / effects, both acute and delayed No information available

## SECTION 12. ECOLOGICAL INFORMATION

## Ecotoxicity effects

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. Do not allow material to contaminate ground water system. The product contains following substances which are hazardous for the environment.

| Component | Freshwater Fish                                                                                                                                                                             | Water Flea                                          | Freshwater Algae | Microtox |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|------------------|----------|
| Silver    | LC50: = 0.064 mg/L,<br>96h static (Lepomis macrochirus)<br>LC50: = 0.0062 mg/L,<br>96h flow-through (Oncorhynchus mykiss)<br>LC50: 0.00155 - 0.00293 mg/L, 96h static (Pimephales promelas) | EC50: = 0.00024 mg/L,<br>48h Static (Daphnia magna) |                  |          |

## Persistence and Degradability

Persistence  
Degradability  
Degradation in sewage treatment plant

Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
Insoluble in water, May persist.  
Not relevant for inorganic substances.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## Bioaccumulative Potential

May have some potential to bioaccumulate; Product has a high potential to bioconcentrate

## Mobility in soil

Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility

Endocrine Disruptor Information  
Persistent Organic Pollutant  
Ozone Depletion Potential

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

## Waste from Residues/Unused Products

Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

## Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

## Other Information

Do not flush to sewer. Do not let this chemical enter the environment. Waste codes should be assigned by the user based on the application for which the product was used. Do not

empty into drains.

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

UN-No UN3077  
Proper Shipping Name Environmentally hazardous substances, solid, n.o.s.  
Technical Shipping Name (Silver)  
Hazard Class 9  
Packing Group III

IMDG/IMO

UN-No UN3077  
Proper Shipping Name Environmentally hazardous substances, solid, n.o.s.  
Technical Shipping Name (Silver)  
Hazard Class 9  
Packing Group III

IATA

UN-No UN3077  
Proper Shipping Name Environmentally hazardous substances, solid, n.o.s.  
Technical Shipping Name (Silver)  
Hazard Class 9  
Packing Group III

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

**International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-----------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Silver    | -                                                   | -                                       | X    | X     | 231-131-3 | X    | X   | X     | X    |      | X    | KE-31261 |

**National Regulations**

## SECTION 16. OTHER INFORMATION

Prepared By Health, Safety and Environmental Department  
Creation Date 05-Apr-2018  
Revision Date 13-May-2024  
Revision Summary New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and

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hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**